2025-11-09-R-Final-Project-V3

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2025-12-03

Library Installation

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(tidyr)
library(stringr)
library(lubridate)
library(readr)
library(ggplot2)
library(dplyr)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(forcats)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
library(tseries)
```

Data Loading

Loading each monthly sales

```r
jan_sales <- read_csv("data/raw/2019-sales-monthly/201901-sales.csv")
```

```
## Rows: 9723 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(jan_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 141234     iPhone                 1                  700          01/22/19 21~
## 2 141235     Lightning Charging Ca~ 1                  14.95        01/28/19 14~
## 3 141236     Wired Headphones       2                  11.99        01/17/19 13~
## 4 141237     27in FHD Monitor       1                  149.99       01/05/19 20~
## 5 141238     Wired Headphones       1                  11.99        01/25/19 11~
## 6 141239     AAA Batteries (4-pack) 1                  2.99         01/29/19 20~
## # i 1 more variable: `Purchase Address` <chr>
```

```r
feb_sales <- read_csv("data/raw/2019-sales-monthly/201902-sales.csv")
```

```
## Rows: 12036 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(feb_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 150502     iPhone                 1                  700          02/18/19 01~
## 2 150503     AA Batteries (4-pack)  1                  3.84         02/13/19 07~
## 3 150504     27in 4K Gaming Monitor 1                  389.99       02/18/19 09~
```

```
## 4 150505     Lightning Charging Ca~ 1                 14.95        02/02/19 16~
## 5 150506     AA Batteries (4-pack)  2                 3.84         02/28/19 20~
## 6 150507     Lightning Charging Ca~ 1                 14.95        02/24/19 18~
## # i 1 more variable: `Purchase Address` <chr>
```

```
mar_sales <- read_csv("data/raw/2019-sales-monthly/201903-sales.csv")
```

```
## Rows: 15226 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(mar_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 162009     iPhone                 1                  700          03/28/19 20~
## 2 162009     Lightning Charging Ca~ 1                  14.95        03/28/19 20~
## 3 162009     Wired Headphones       2                  11.99        03/28/19 20~
## 4 162010     Bose SoundSport Headp~ 1                  99.99        03/17/19 05~
## 5 162011     34in Ultrawide Monitor 1                  379.99       03/10/19 00~
## 6 162012     AA Batteries (4-pack)  1                  3.84         03/20/19 21~
## # i 1 more variable: `Purchase Address` <chr>
```

```
apr_sales <- read_csv("data/raw/2019-sales-monthly/201904-sales.csv")
```

```
## Rows: 18383 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(apr_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 176558     USB-C Charging Cable   2                  11.95        04/19/19 08~
## 2 <NA>       <NA>                   <NA>               <NA>         <NA>
## 3 176559     Bose SoundSport Headp~ 1                  99.99        04/07/19 22~
## 4 176560     Google Phone           1                  600          04/12/19 14~
## 5 176560     Wired Headphones       1                  11.99        04/12/19 14~
## 6 176561     Wired Headphones       1                  11.99        04/30/19 09~
## # i 1 more variable: `Purchase Address` <chr>
```

```r
may_sales <- read_csv("data/raw/2019-sales-monthly/201905-sales.csv")
```

```
## Rows: 16635 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(may_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product               `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                 <chr>              <chr>        <chr>
## 1 194095     Wired Headphones      1                  11.99        05/16/19 17~
## 2 194096     AA Batteries (4-pack) 1                  3.84         05/19/19 14~
## 3 194097     27in FHD Monitor      1                  149.99       05/24/19 11~
## 4 194098     Wired Headphones      1                  11.99        05/02/19 20~
## 5 194099     AAA Batteries (4-pack) 2                 2.99         05/11/19 22~
## 6 194100     iPhone                1                  700.0        05/10/19 19~
## # i 1 more variable: `Purchase Address` <chr>
```

```r
jun_sales <- read_csv("data/raw/2019-sales-monthly/201906-sales.csv")
```

```
## Rows: 13622 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(jun_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product               `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                 <chr>              <chr>        <chr>
## 1 209921     USB-C Charging Cable  1                  11.95        06/23/19 19~
## 2 209922     Macbook Pro Laptop    1                  1700.0       06/30/19 10~
## 3 209923     ThinkPad Laptop       1                  999.99       06/24/19 20~
## 4 209924     27in FHD Monitor      1                  149.99       06/05/19 10~
## 5 209925     Bose SoundSport Headp~ 1                 99.99        06/25/19 18~
## 6 209926     Apple Airpods Headpho~ 1                 150.0        06/28/19 20~
## # i 1 more variable: `Purchase Address` <chr>
```

```r
jul_sales <- read_csv("data/raw/2019-sales-monthly/201907-sales.csv")
```

```
## Rows: 14371 Columns: 6
## -- Column specification --------------------------------------------------------
```

```
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(jul_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 222910     Apple Airpods Headpho~ 1                  150          07/26/19 16~
## 2 222911     Flatscreen TV          1                  300          07/05/19 08~
## 3 222912     AA Batteries (4-pack)  1                  3.84         07/29/19 12~
## 4 222913     AA Batteries (4-pack)  1                  3.84         07/28/19 10~
## 5 222914     AAA Batteries (4-pack) 5                  2.99         07/31/19 02~
## 6 222915     Bose SoundSport Headp~ 1                  99.99        07/03/19 18~
## # i 1 more variable: `Purchase Address` <chr>
```

```
aug_sales <- read_csv("data/raw/2019-sales-monthly/201908-sales.csv")
```

```
## Rows: 12011 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(aug_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 236670     Wired Headphones       2                  11.99        08/31/19 22~
## 2 236671     Bose SoundSport Headp~ 1                  99.99        08/15/19 15~
## 3 236672     iPhone                 1                  700.0        08/06/19 14~
## 4 236673     AA Batteries (4-pack)  2                  3.84         08/29/19 20~
## 5 236674     AA Batteries (4-pack)  2                  3.84         08/15/19 19~
## 6 236675     Wired Headphones       1                  11.99        08/02/19 23~
## # i 1 more variable: `Purchase Address` <chr>
```

```
sep_sales <- read_csv("data/raw/2019-sales-monthly/201909-sales.csv")
```

```
## Rows: 11686 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(sep_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 248151     AA Batteries (4-pack)  4                  3.84         09/17/19 14~
## 2 248152     USB-C Charging Cable   2                  11.95        09/29/19 10~
## 3 248153     USB-C Charging Cable   1                  11.95        09/16/19 17~
## 4 248154     27in FHD Monitor       1                  149.99       09/27/19 07~
## 5 248155     USB-C Charging Cable   1                  11.95        09/01/19 19~
## 6 248156     34in Ultrawide Monitor 1                  379.99       09/13/19 14~
## # i 1 more variable: `Purchase Address` <chr>
```

```r
oct_sales <- read_csv("data/raw/2019-sales-monthly/201910-sales.csv")
```

```
## Rows: 20379 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(oct_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 259358     34in Ultrawide Monitor 1                  379.99       10/28/19 10~
## 2 259359     27in 4K Gaming Monitor 1                  389.99       10/28/19 17~
## 3 259360     AAA Batteries (4-pack) 2                  2.99         10/24/19 17~
## 4 259361     27in FHD Monitor       1                  149.99       10/14/19 22~
## 5 259362     Wired Headphones       1                  11.99        10/07/19 16~
## 6 259363     AAA Batteries (4-pack) 1                  2.99         10/01/19 18~
## # i 1 more variable: `Purchase Address` <chr>
```

```r
nov_sales <- read_csv("data/raw/2019-sales-monthly/201911-sales.csv")
```

```
## Rows: 17661 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
head(nov_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product                `Quantity Ordered` `Price Each` `Order Date`
```

```
##   <chr>      <chr>                  <chr>              <chr>        <chr>
## 1 278797     Wired Headphones       1                  11.99        11/21/19 09~
## 2 278798     USB-C Charging Cable   2                  11.95        11/17/19 10~
## 3 278799     Apple Airpods Headpho~ 1                  150.0        11/19/19 14~
## 4 278800     27in FHD Monitor       1                  149.99       11/25/19 22~
## 5 278801     Bose SoundSport Headp~ 1                  99.99        11/09/19 13~
## 6 278802     USB-C Charging Cable   1                  11.95        11/14/19 20~
## # i 1 more variable: `Purchase Address` <chr>
```

```
dec_sales <- read_csv("data/raw/2019-sales-monthly/201912-sales.csv")
```

```
## Rows: 25117 Columns: 6
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (6): Order ID, Product, Quantity Ordered, Price Each, Order Date, Purcha...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(dec_sales)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product               `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                 <chr>              <chr>        <chr>
## 1 295665     Macbook Pro Laptop    1                  1700         12/30/19 00:~
## 2 295666     LG Washing Machine    1                  600.0        12/29/19 07:~
## 3 295667     USB-C Charging Cable  1                  11.95        12/12/19 18:~
## 4 295668     27in FHD Monitor      1                  149.99       12/22/19 15:~
## 5 295669     USB-C Charging Cable  1                  11.95        12/18/19 12:~
## 6 295670     AA Batteries (4-pack) 1                  3.84         12/31/19 22:~
## # i 1 more variable: `Purchase Address` <chr>
```

Combine the 12 months data using bind_rows

```
all_sales_2019_combine <- bind_rows(jan_sales, feb_sales, mar_sales,
                            apr_sales, may_sales, jun_sales,
                            jul_sales, aug_sales, sep_sales,
                            oct_sales, nov_sales, dec_sales)

head(all_sales_2019_combine)
```

```
## # A tibble: 6 x 6
##   `Order ID` Product               `Quantity Ordered` `Price Each` `Order Date`
##   <chr>      <chr>                 <chr>              <chr>        <chr>
## 1 141234     iPhone                1                  700          01/22/19 21~
## 2 141235     Lightning Charging Ca~ 1                  14.95        01/28/19 14~
## 3 141236     Wired Headphones       2                  11.99        01/17/19 13~
## 4 141237     27in FHD Monitor       1                  149.99       01/05/19 20~
## 5 141238     Wired Headphones       1                  11.99        01/25/19 11~
## 6 141239     AAA Batteries (4-pack) 1                  2.99         01/29/19 20~
## # i 1 more variable: `Purchase Address` <chr>
```

EDA Process

Clean the Data

Check the Table Info

```
glimpse(all_sales_2019_combine)
```

```
## Rows: 186,850
## Columns: 6
## $ `Order ID`         <chr> "141234", "141235", "141236", "141237", "141238", "~
## $ Product            <chr> "iPhone", "Lightning Charging Cable", "Wired Headph~
## $ `Quantity Ordered` <chr> "1", "1", "2", "1", "1", "1", "1", "1", "1", "1", "~
## $ `Price Each`       <chr> "700", "14.95", "11.99", "149.99", "11.99", "2.99",~
## $ `Order Date`       <chr> "01/22/19 21:25", "01/28/19 14:15", "01/17/19 13:33~
## $ `Purchase Address` <chr> "944 Walnut St, Boston, MA 02215", "185 Maple St, P~
```

Counts the Number of Missing Cells (NA)

```
total_NA_cells <- sum(is.na(all_sales_2019_combine))
cat("Total no. of missing cells: ", total_NA_cells, "\n")
```

```
## Total no. of missing cells:  3270
```

Counts the Number of Rows with at Least One NA

```
total_NA_rows <- sum(!complete.cases(all_sales_2019_combine))
cat("Total No. of missing rows: ", total_NA_rows, "\n")
```

```
## Total No. of missing rows:  545
```

Counting Total No. of Rows Before Dropping

```
row_before <- nrow(all_sales_2019_combine)
cat("Number of rows before dropping NA: ", row_before, "\n")
```

```
## Number of rows before dropping NA:  186850
```

Dropping NA Rows

```
all_sales_2019 <- drop_na(all_sales_2019_combine)
# counting total no. of rows after dropping
row_after <- nrow(all_sales_2019)
cat("Number of rows after dropping NA: ", row_after, "\n")
```

```
## Number of rows after dropping NA:  186305
```

```
# if this match with the total no. of missing row, we can validate it
cat("No. of dropped rows: ", row_before - row_after)
```

```
## No. of dropped rows:  545
```

Removing Duplicated Header Rows

```
all_sales_2019 <- all_sales_2019 %>%
  # this remove the no. of repeated header rows
  filter(`Order ID` != "Order ID") %>%
  mutate(
    OrderID = `Order ID`,
    QuantityOrdered = as.integer(`Quantity Ordered`),
    PriceEach = as.numeric(`Price Each`),
    OrderDate = `Order Date`,
    PurchaseAddress = `Purchase Address`,
    ZIP = str_extract(PurchaseAddress, "\\d{5}$")
  ) %>%
  select(OrderID, Product, QuantityOrdered, PriceEach, OrderDate, PurchaseAddress, ZIP)

row_after_cleaning_repeated_header <- nrow(all_sales_2019)

cat("Total no. of repeated header rows that was dropped: ", row_after - row_after_cleaning_repeated_hea
```

```
## Total no. of repeated header rows that was dropped:  355
```

Check the Table Info & Exporting

```
glimpse(all_sales_2019)
```

```
## Rows: 185,950
## Columns: 7
## $ OrderID         <chr> "141234", "141235", "141236", "141237", "141238", "141~
## $ Product         <chr> "iPhone", "Lightning Charging Cable", "Wired Headphone~
## $ QuantityOrdered <int> 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, ~
## $ PriceEach       <dbl> 700.00, 14.95, 11.99, 149.99, 11.99, 2.99, 389.99, 11.~
## $ OrderDate       <chr> "01/22/19 21:25", "01/28/19 14:15", "01/17/19 13:33", ~
## $ PurchaseAddress <chr> "944 Walnut St, Boston, MA 02215", "185 Maple St, Port~
## $ ZIP             <chr> "02215", "97035", "94016", "90001", "73301", "94016", ~
```

```
write_csv(all_sales_2019, "data/processed/all_sales_2019.csv")
```

Data Manipulation

Separating PurchaseAddress into 3 Columns (Street, City and State_ZIP)

```r
all_sales_2019_splitted_and_formatted <- separate(all_sales_2019, PurchaseAddress, into = c("Street", "
```

Add TotalSales & Separate City and ZIP columns

```r
all_sales_2019_splitted_and_formatted <- all_sales_2019_splitted_and_formatted %>%
  mutate(
    TotalSales = QuantityOrdered * PriceEach,
    City = str_c(City, " (", str_extract(State_ZIP, "^[A-Z]{2}"), ")"),
    ZIP = str_extract(State_ZIP, "\\d{5}$")
    )
```

Adding OrderDataTime, Month, and Hour Columns

```r
all_sales_2019_splitted_and_formatted <- all_sales_2019_splitted_and_formatted %>%
  mutate(
    OrderDateTime = mdy_hm(OrderDate),
    Month = month(OrderDateTime, label = TRUE),
    Hour = hour(OrderDateTime)
  )
```

Select and Check the Clean Data

```r
all_sales_2019_splitted_and_formatted <- all_sales_2019_splitted_and_formatted %>%
  select(
    OrderID, Product, QuantityOrdered, PriceEach, TotalSales, Street, City, ZIP, Month, Hour, OrderDate
    )

glimpse(all_sales_2019_splitted_and_formatted)
```

```
## Rows: 185,950
## Columns: 11
## $ OrderID         <chr> "141234", "141235", "141236", "141237", "141238", "141~
## $ Product         <chr> "iPhone", "Lightning Charging Cable", "Wired Headphone~
## $ QuantityOrdered <int> 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, ~
## $ PriceEach       <dbl> 700.00, 14.95, 11.99, 149.99, 11.99, 2.99, 389.99, 11.~
## $ TotalSales      <dbl> 700.00, 14.95, 23.98, 149.99, 11.99, 2.99, 389.99, 11.~
## $ Street          <chr> "944 Walnut St", "185 Maple St", "538 Adams St", "738 ~
## $ City            <chr> "Boston (MA)", "Portland (OR)", "San Francisco (CA)", ~
## $ ZIP             <chr> "02215", "97035", "94016", "90001", "73301", "94016", ~
## $ Month           <ord> Jan, Jan, Jan, Jan, Jan, Jan, Jan, Jan, Jan, Jan, Jan,~
## $ Hour            <int> 21, 14, 13, 20, 11, 20, 12, 12, 10, 21, 11, 10, 18, 19~
## $ OrderDateTime   <dttm> 2019-01-22 21:25:00, 2019-01-28 14:15:00, 2019-01-17 ~
```

Exporting Clean Data into CSV

```
# write_csv(all_sales_2019_splitted_and_formatted, "data/processed/all_sales_2019_v2.csv")
```

EDA Process

Importing Data

```
all_sales_2019 <- read_csv("data/processed/all_sales_2019_v2.csv")
```

```
## Rows: 185950 Columns: 11
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (5): Product, Street, City, ZIP, Month
## dbl  (5): OrderID, QuantityOrdered, PriceEach, TotalSales, Hour
## dttm (1): OrderDateTime
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# Check data structure and Null Value.
head(all_sales_2019)
```

```
## # A tibble: 6 x 11
##   OrderID Product  QuantityOrdered PriceEach TotalSales Street City  ZIP   Month
##     <dbl> <chr>              <dbl>     <dbl>      <dbl> <chr>  <chr> <chr> <chr>
## 1  141234 iPhone                 1    700        700    944 W~ Bost~ 02215 Jan
## 2  141235 Lightni~               1     15.0       15.0  185 M~ Port~ 97035 Jan
## 3  141236 Wired H~               2     12.0       24.0  538 A~ San ~ 94016 Jan
## 4  141237 27in FH~               1    150.       150.   738 1~ Los ~ 90001 Jan
## 5  141238 Wired H~               1     12.0       12.0  387 1~ Aust~ 73301 Jan
## 6  141239 AAA Bat~               1      2.99       2.99 775 W~ San ~ 94016 Jan
## # i 2 more variables: Hour <dbl>, OrderDateTime <dttm>
```

```
glimpse(all_sales_2019)
```

```
## Rows: 185,950
## Columns: 11
## $ OrderID         <dbl> 141234, 141235, 141236, 141237, 141238, 141239, 141240~
## $ Product         <chr> "iPhone", "Lightning Charging Cable", "Wired Headphone~
## $ QuantityOrdered <dbl> 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, ~
## $ PriceEach       <dbl> 700.00, 14.95, 11.99, 149.99, 11.99, 2.99, 389.99, 11.~
## $ TotalSales      <dbl> 700.00, 14.95, 23.98, 149.99, 11.99, 2.99, 389.99, 11.~
## $ Street          <chr> "944 Walnut St", "185 Maple St", "538 Adams St", "738 ~
## $ City            <chr> "Boston (MA)", "Portland (OR)", "San Francisco (CA)", ~
## $ ZIP             <chr> "02215", "97035", "94016", "90001", "73301", "94016", ~
## $ Month           <chr> "Jan", "Jan", "Jan", "Jan", "Jan", "Jan", "Jan", "Jan"~
## $ Hour            <dbl> 21, 14, 13, 20, 11, 20, 12, 12, 10, 21, 11, 10, 18, 19~
## $ OrderDateTime   <dttm> 2019-01-22 21:25:00, 2019-01-28 14:15:00, 2019-01-17 ~
```

Calcuating Summary Statistics

```
cat("=== General Summary of Data ===\n")
```

```
## === General Summary of Data ===
```

```
summary(all_sales_2019 %>% select(QuantityOrdered, PriceEach, TotalSales))
```

```
##  QuantityOrdered   PriceEach         TotalSales
##  Min.   :1.000   Min.   :   2.99   Min.   :   2.99
##  1st Qu.:1.000   1st Qu.:  11.95   1st Qu.:  11.95
##  Median :1.000   Median :  14.95   Median :  14.95
##  Mean   :1.124   Mean   : 184.40   Mean   : 185.49
##  3rd Qu.:1.000   3rd Qu.: 150.00   3rd Qu.: 150.00
##  Max.   :9.000   Max.   :1700.00   Max.   :3400.00
```

Checking Null Value

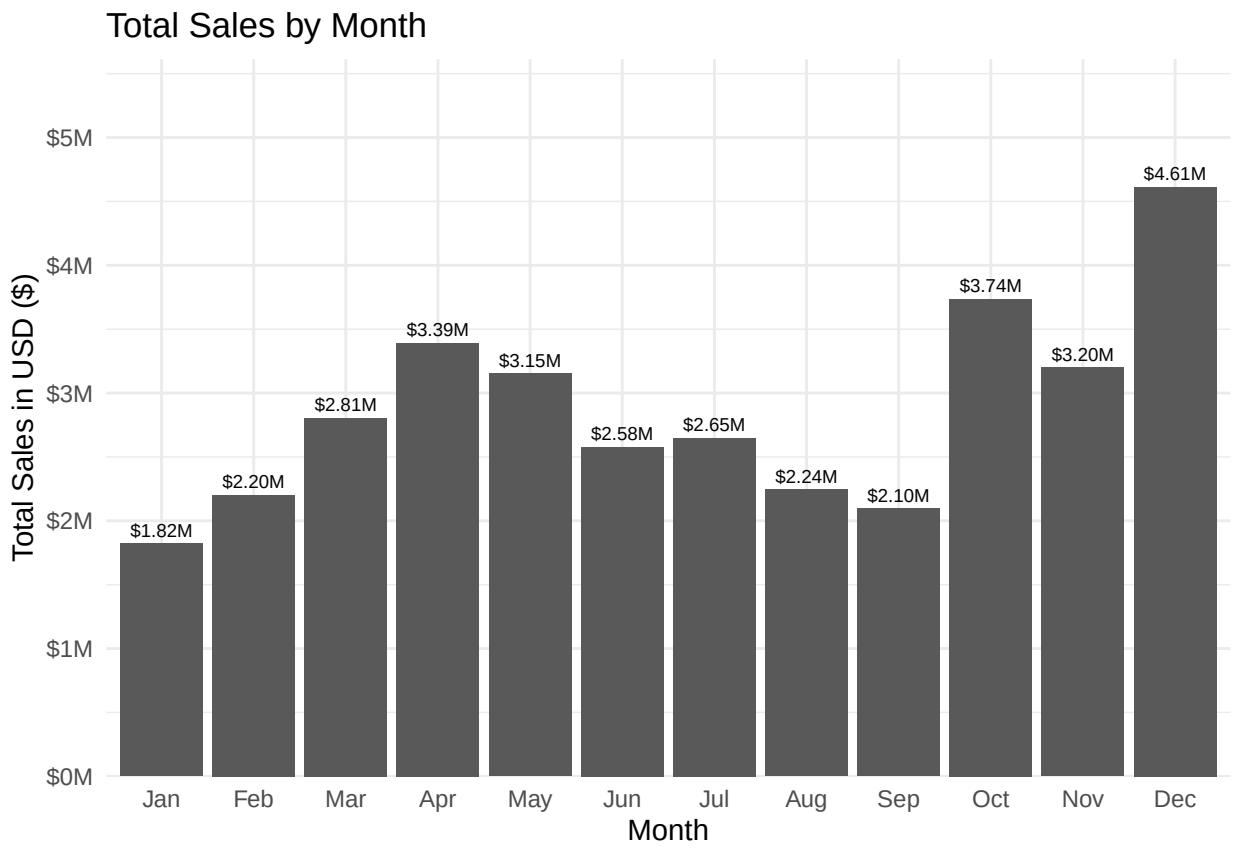
```
sum(is.na(all_sales_2019))
```

```
## [1] 0
```

Question 1: What was the best month for sales? How much was earned that month?

```
total_sales_each_month <- all_sales_2019 %>%
  group_by(Month) %>%
  summarize(TotalSales = sum(TotalSales)) %>%
  arrange(desc(TotalSales))

# write_csv(total_sales_each_month, "outputs/tables/total_sales_each_month.csv")

month_order = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
                "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

# ordering month for visual
total_sales_each_month <- total_sales_each_month %>%
  mutate(Month = factor(Month, levels = month_order))

vis_1 <- ggplot(total_sales_each_month, aes(x = Month, y = TotalSales)) +
  geom_col() +
  labs(title = "Total Sales by Month",
       x = "Month",
       y = "Total Sales in USD ($)")

vis_1 <- vis_1 +
```

```r
  geom_text(aes(label = scales::dollar(TotalSales, scale = 1/1e6, suffix = "M")),
            vjust = -0.5,
            size = 2.5) +
  scale_y_continuous(
    labels = scales::dollar_format(scale = 1/1e6, suffix = "M"),
    expand = expansion(add = c(0, 1e6))
    ) +
  theme_minimal()

# ggsave("outputs/figures/total_sales_by_month.jpg", plot = vis_1)
print(vis_1)
```

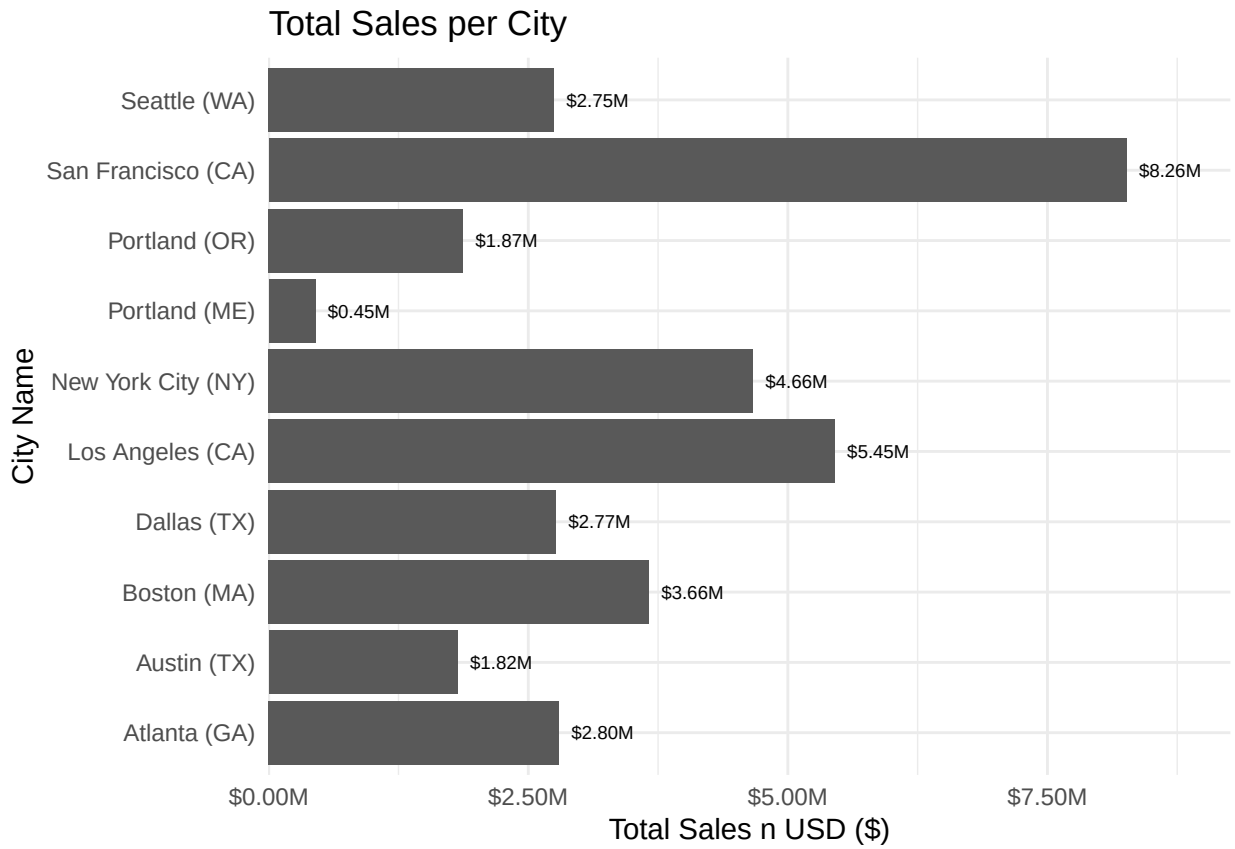



```r
best_month_for_sales <- total_sales_each_month %>%
  slice_max(TotalSales, n = 1)

cat("The best month for sales is", best_month_for_sales$Month,
    "with the total sales of",
    paste0("$", best_month_for_sales$TotalSales, "."),
    "\n")
```

```
## The best month for sales is 12 with the total sales of $4613443.34.
```

Question 2: What city has the highest sales?

```r
total_sales_each_city <- all_sales_2019 %>%
  group_by(City) %>%
  summarize(TotalSales = sum(TotalSales)) %>%
  arrange(desc(TotalSales))

# write_csv(total_sales_each_city, "outputs/tables/total_sales_each_city.csv")

vis_2 <- ggplot(total_sales_each_city, aes(TotalSales, City)) +
  geom_col()

vis_2 <- vis_2 +
  labs(
    title = "Total Sales per City",
    x = "Total Sales n USD ($)",
    y = "City Name"
  )+
  geom_text(aes(label = scales::dollar(TotalSales, scale = 1/1e6, suffix = "M")),
                  hjust = -0.2,
                  size = 2.5) +
  scale_x_continuous(
    label = scales::dollar_format(scale = 1/1e6, suffix = "M"),
     expand = expansion(add = c(0, 1e6))
    ) +
  theme_minimal()

# ggsave("outputs/figures/total_sales_per_city.jpg", plot = vis_2)
print(vis_2)
```




```
city_with_highest_sales <- total_sales_each_city %>%
  slice_max(TotalSales, n = 1)


cat("The city with highest sales is", city_with_highest_sales$City,
    "with the total sales of",
    paste0("$", city_with_highest_sales$TotalSales, "."),
    "\n")
```

```
## The city with highest sales is San Francisco (CA) with the total sales of $8262203.91.
```

Question 3: What time should we display advertisements to maximize likehood of customers buying products?

```
glimpse(all_sales_2019)
```

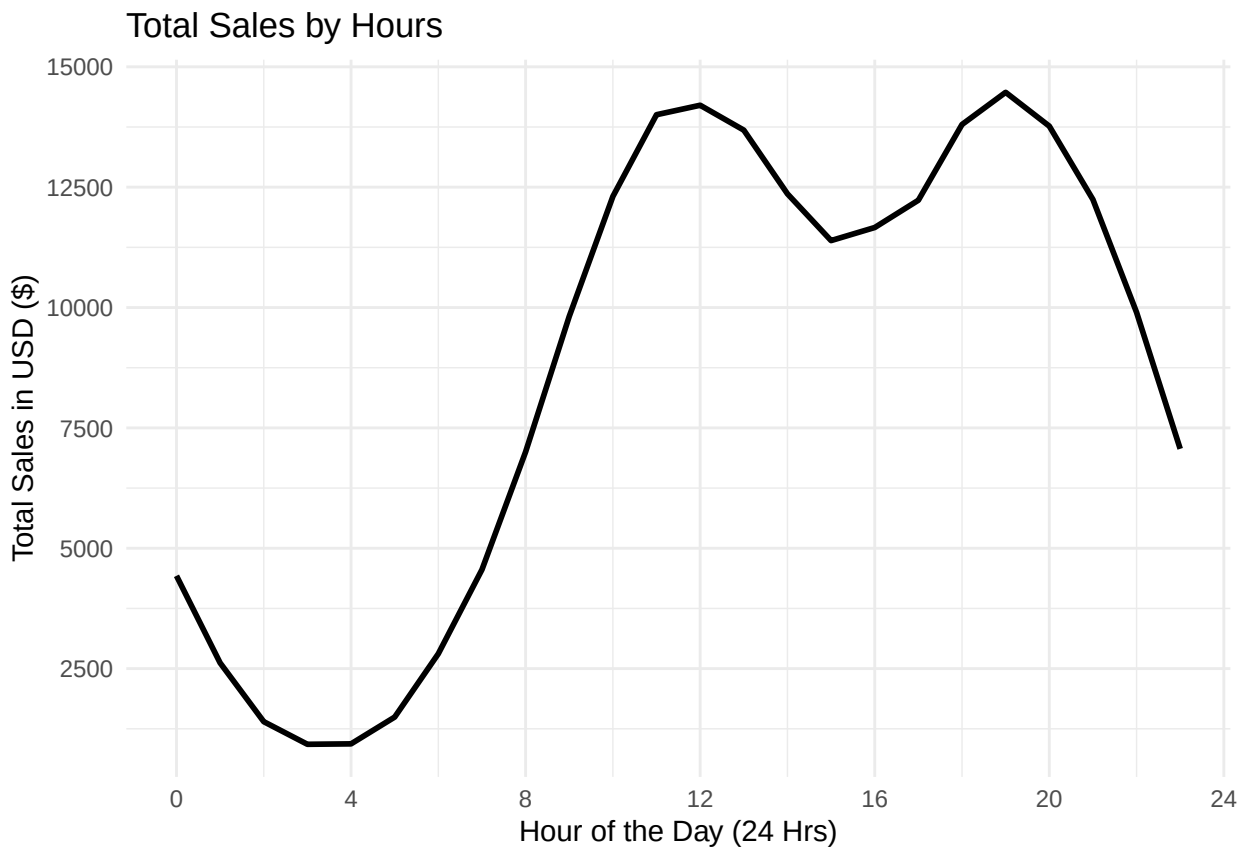
```
## Rows: 185,950
## Columns: 11
## $ OrderID         <dbl> 141234, 141235, 141236, 141237, 141238, 141239, 141240~
## $ Product         <chr> "iPhone", "Lightning Charging Cable", "Wired Headphone~
## $ QuantityOrdered <dbl> 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 3, 1, 1, 1, 1, 1, ~
## $ PriceEach       <dbl> 700.00, 14.95, 11.99, 149.99, 11.99, 2.99, 389.99, 11.~
```

```
## $ TotalSales      <dbl> 700.00, 14.95, 23.98, 149.99, 11.99, 2.99, 389.99, 11.~
## $ Street          <chr> "944 Walnut St", "185 Maple St", "538 Adams St", "738 ~
## $ City            <chr> "Boston (MA)", "Portland (OR)", "San Francisco (CA)", ~
## $ ZIP             <chr> "02215", "97035", "94016", "90001", "73301", "94016", ~
## $ Month           <chr> "Jan", "Jan", "Jan", "Jan", "Jan", "Jan", "Jan", "Jan"~
## $ Hour            <dbl> 21, 14, 13, 20, 11, 20, 12, 12, 10, 21, 11, 10, 18, 19~
## $ OrderDateTime   <dttm> 2019-01-22 21:25:00, 2019-01-28 14:15:00, 2019-01-17 ~
```

```r
sales_by_hours <- all_sales_2019 %>%
  group_by(Hour) %>%
  summarize(TotalQuantitySales = sum(QuantityOrdered))

# write_csv(sales_by_hours, "outputs/tables/sales_by_hours.csv")

vis_3 <- ggplot(sales_by_hours, aes(x = Hour, y = TotalQuantitySales)) +
  geom_line(size = 1) +
  labs(
    title = "Total Sales by Hours",
    y = "Total Sales in USD ($)",
    x = "Hour of the Day (24 Hrs)"
  ) +
  scale_x_continuous(breaks = seq(0, 24, by = 4)) +
  scale_y_continuous(breaks = seq(0, 17500, by = 2500)) +
  theme_minimal()
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```r
# ggsave("outputs/figures/total_sales_by_hours.jpg", plot = vis_3)
print(vis_3)
```



Question 4: What products are most often sold together?

```
duplicated_OrderID <- all_sales_2019 %>%
  group_by(OrderID) %>%
  filter(n() > 1) %>%
  select(OrderID, Product) %>%
  ungroup()

products_sold_together <- duplicated_OrderID %>%
  group_by(OrderID) %>%
#  count(sort = TRUE) %>%
  summarize(ProductPairs = paste(sort(Product), collapse = ", ")) %>%
  ungroup()

products_sold_together_count <- products_sold_together %>%
  count(ProductPairs, sort = TRUE) %>%
  rename(TotalSales = n)

top_10_products_sold_together <- head(products_sold_together_count, n = 10)

print(top_10_products_sold_together)
```

```
## # A tibble: 10 x 2
```

```
##    ProductPairs                                    TotalSales
##    <chr>                                                <int>
##  1 iPhone, Lightning Charging Cable                       891
##  2 Google Phone, USB-C Charging Cable                     868
##  3 iPhone, Wired Headphones                               374
##  4 USB-C Charging Cable, Vareebadd Phone                  318
##  5 Google Phone, Wired Headphones                         311
##  6 Apple Airpods Headphones, iPhone                       299
##  7 Bose SoundSport Headphones, Google Phone               169
##  8 Vareebadd Phone, Wired Headphones                      110
##  9 AA Batteries (4-pack), Lightning Charging Cable        103
## 10 Lightning Charging Cable, USB-C Charging Cable          96
```

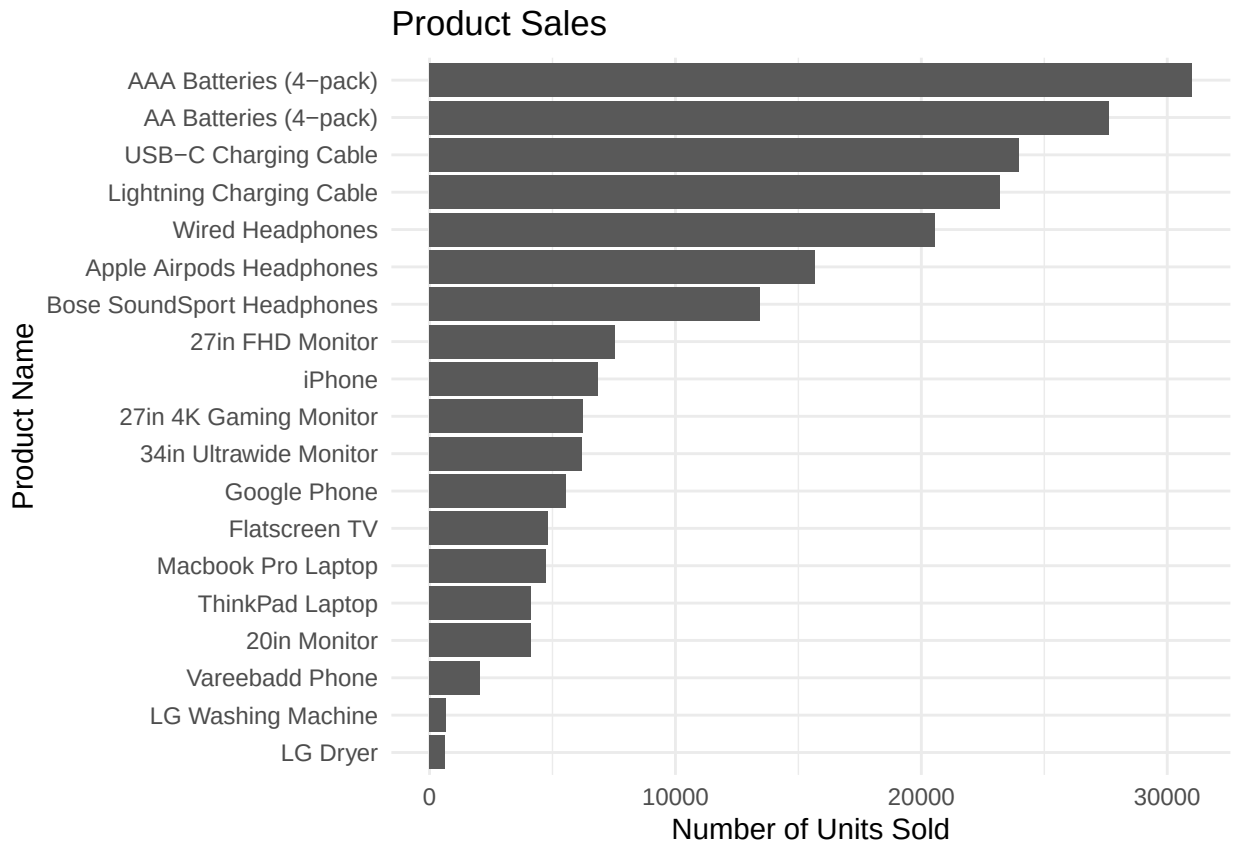
```
# write_csv(products_sold_together_count, "outputs/tables/products_sold_together.csv")
```

Question 5: What product sold the most? Why?

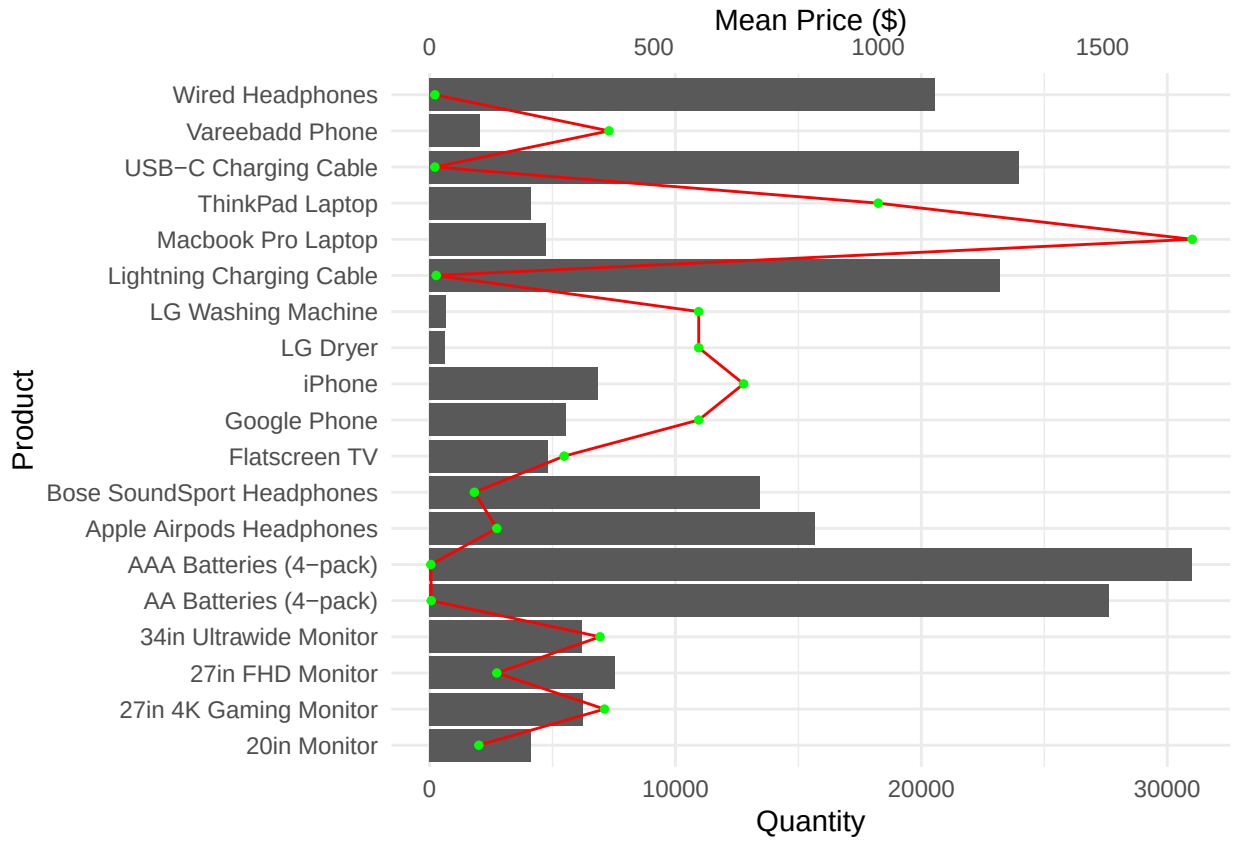
```
single_product_sold <- all_sales_2019 %>%
  group_by(Product) %>%
  count(wt = QuantityOrdered, sort = TRUE) %>%
  rename(TotalUnitSold = n)

most_single_product_sold <- single_product_sold %>%
  head(1)


cat(paste0("The most sold product is ", most_single_product_sold$Product,
           " with the total sales of ", most_single_product_sold$TotalUnitSold,
           " items. \n"))
```

```
## The most sold product is AAA Batteries (4-pack) with the total sales of 31017 items.
```

```
# write_csv(single_product_sold, "outputs/tables/product_sales.csv")

vis_4 <- ggplot(single_product_sold, aes(TotalUnitSold, fct_reorder(Product, TotalUnitSold))) +
  geom_col() +
  labs(
    title = "Product Sales",
    x = "Number of Units Sold",
    y = "Product Name"
  ) +
  theme_minimal()

# ggsave("outputs/figures/product_sales.jpg", plot = vis_4)
print(vis_4)
```




```r
# Get Quantity and Price by Product
trend <- all_sales_2019 %>%
  group_by(Product) %>%
  summarize(Qty = sum(QuantityOrdered), Price = mean(PriceEach))

# Calculate scaling factor
coeff <- max(trend$Qty) / max(trend$Price)

#
ggplot(trend, aes(x = Product)) +
  geom_col(aes(y = Qty)) +
  geom_line(aes(y = Price * coeff, group = 1), color = "red", size = 0.5) +
  geom_point(aes(y = Price * coeff), color = "green", size = 1) +
  scale_y_continuous(
    name = "Quantity",
    sec.axis = sec_axis(~ . / coeff, name = "Mean Price ($)")
  ) +
  coord_flip() +  # rotates the chart
  theme_minimal()
```



From the above viausl we can see **inverse correlation**, which is cheap products (like AAA Batteries and Charging Cables) sell in very high quantities, while expensive products (like Laptops and Monitors) sell in lower quantities. However, there are some exception such as Even though the Mac book Pro Laptop is the most expensive item (the green dot is furthest to the right), it still sells more units than cheaper home appliances like the LG Washing Machine. This shows that customer demand for laptops is much higher than for appliances, regardless of the high price.

This proves that while low price usually drives high volume, strong brand popularity (like Apple) or high necessity (like work laptops) can still drive sales even at high prices.

Building Time series model

```
# all_sales_2019 <- read_csv("data/processed/all_sales_2019_v2.csv")
```

aggregate sales by day

```
daily_sales <- all_sales_2019 %>%
  mutate(Date = as.Date(OrderDateTime)) %>% # Date format
  group_by(Date) %>%
  summarize(TotalSales = sum(TotalSales)) %>%
  arrange(Date) # data is ordered chronologically
```

create a Time Series Object

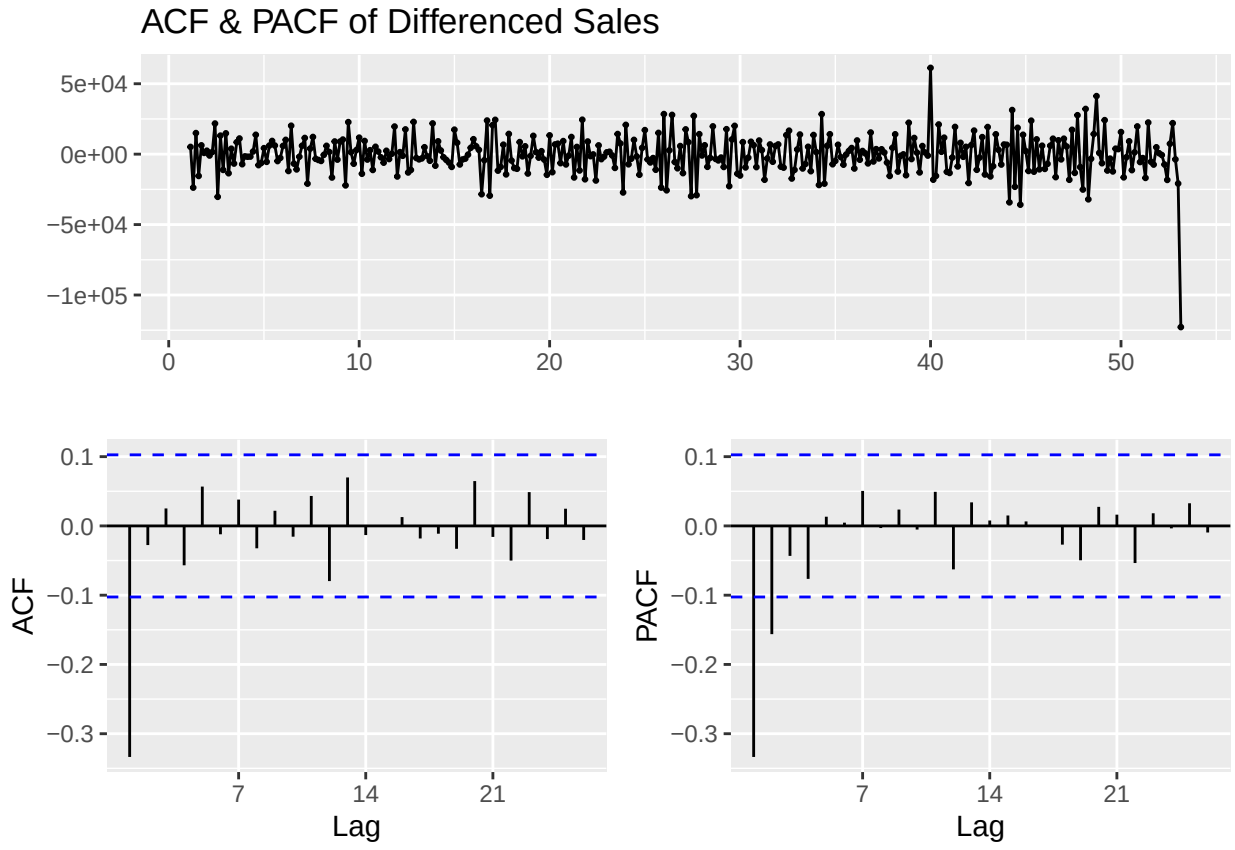
```
# use frequency = 7 >>> weekly seasonality
sales_ts2 <- ts(daily_sales$TotalSales, frequency = 7)
```

analyze Differencing

```
sales_diff <- diff(sales_ts2)
# we look at the Differenced data (Day 2 - Day 1)
```

plot ACF and PACF

```
ggtsdisplay(sales_diff, main="ACF & PACF of Differenced Sales")
```



The ACF plot shows clear spikes at Lag 7, 14, and 21. This confirm a strong weekly cycle, meaning sales consistently peak on specific days (likely weekends). The top chart shows the data fluctuating around zero without drifting up or down, which means the data is stationary after differencing.

MODEL BUILDING

Model A: (ARIMA 0,1,1)

```
model_a <- Arima(sales_ts2, order = c(0,1,1))
```

Model B: (ARIMA 1,1,1)

```
model_b <- Arima(sales_ts2, order = c(1,1,1))
# checks if yesterday's sales value directly predicts today's
```

Model C: (SARIMA 1,1,1)(1,0,0)[7]

```
model_c <- Arima(sales_ts2,
                 order = c(1,1,1),
                 seasonal = list(order = c(1,1,1), period = 7))
# looks for a 7-day (weekly) pattern
```

Compare Models

```
cat("AIC Comparison (note: Lower is Better):\n")
```

```
## AIC Comparison (note: Lower is Better):
```

```
cat("Model A (0,1,1):", AIC(model_a), "\n")
```

```
## Model A (0,1,1): 7950.204
```

```
cat("Model B (1,1,1):", AIC(model_b), "\n")
```

```
## Model B (1,1,1): 7952.06
```

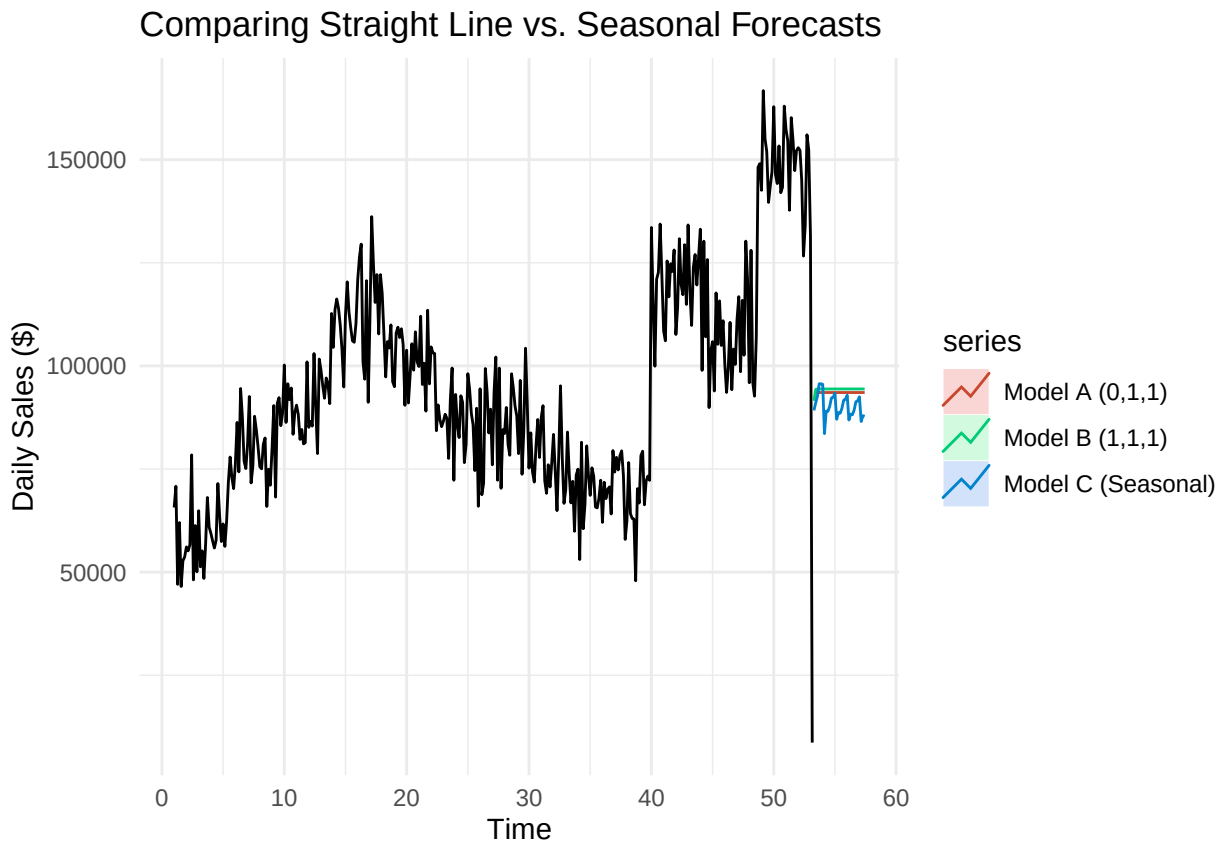
```
cat("Model C (Seasonal):", AIC(model_c), "\n")
```

```
## Model C (Seasonal): 7833.535
```

We tested three models and Model C (Seasonal ARIMA) was the most accurate because it explicitly account for the weekly sales pattern. It achieve the lowest AIC score (7833.5), which is significantly better than the non-seasonal models (7950.2 and 7952.0), indicating much tighter fit to the real data.

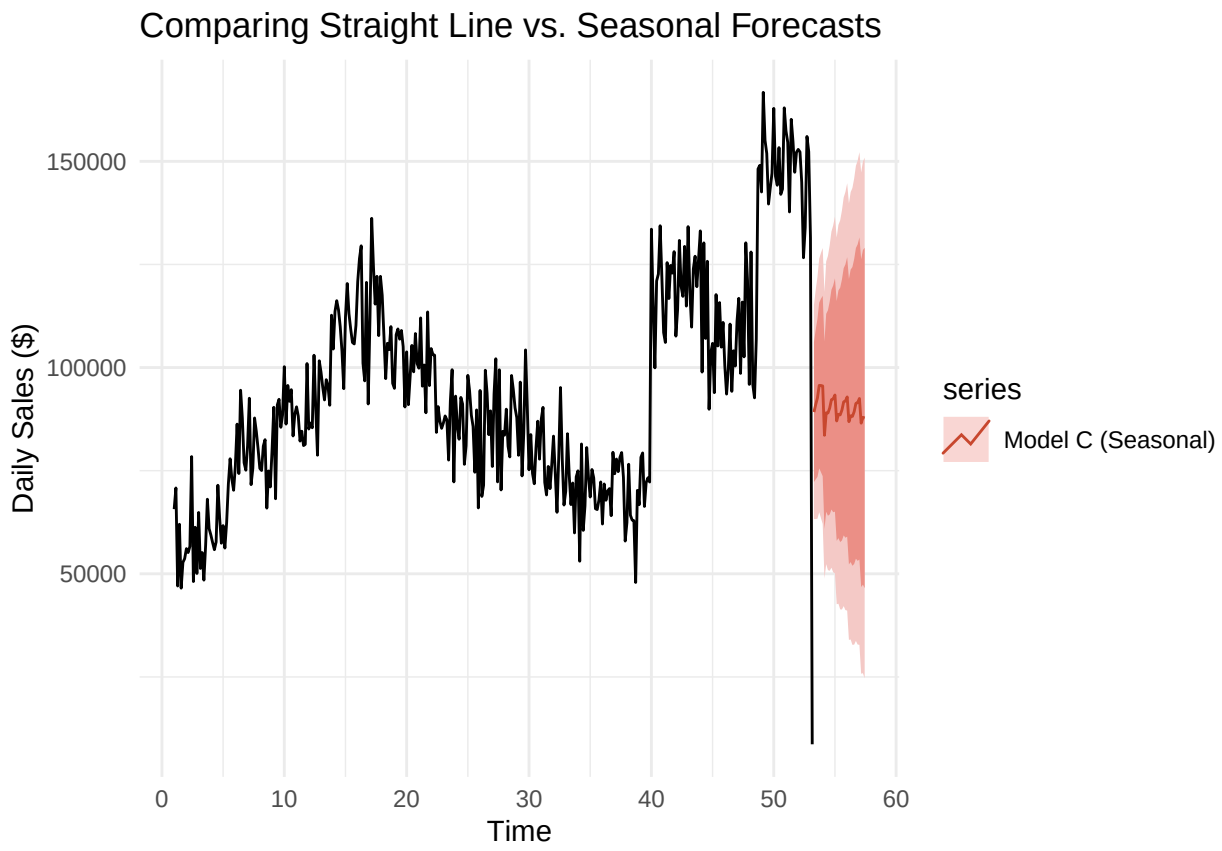
Visual Comparison of Forecasts

```
autoplot(sales_ts2) +
  autolayer(forecast(model_a, h=30), series="Model A (0,1,1)", PI=FALSE) +
  autolayer(forecast(model_b, h=30), series="Model B (1,1,1)", PI=FALSE) +
  autolayer(forecast(model_c, h=30), series="Model C (Seasonal)", PI=FALSE) +
  labs(title = "Comparing Straight Line vs. Seasonal Forecasts",
       y = "Daily Sales ($)") +
  theme_minimal()
```



visualization with the uncertainty

```
autoplot(sales_ts2) +
  autolayer(forecast(model_c, h=30), series="Model C (Seasonal)", PI=TRUE) +
  labs(title = "Comparing Straight Line vs. Seasonal Forecasts",
       y = "Daily Sales ($)") +
  theme_minimal()
```



Unlike simple straight line, our final forecast predict that sales will continue to fluctuate week-by-week, preserving the historical "up and down" trend of the business. This is more realistic than the simple linear regression we previously tested, which only explained about 27% of the data variation.

Building Linear Model (Archive Part - only for review purpose)

Please note that we only add this part to show what we have done before Professor feedback during the presentation. We have changed our model to Time Series as mentioned above according to professor feedback

1. Prepare Data: Aggregate sales by day

```
daily_sales2 <- all_sales_2019 %>%
  mutate(Date = as.Date(OrderDateTime, format = "%m/%d/%y %H:%M")) %>%
  group_by(Date) %>%
  summarize(TotalSales = sum(QuantityOrdered * PriceEach)) %>%
  mutate(DayIndex = as.numeric(Date - min(Date) + 1)) # Create a numeric time variable
```

2. Build Linear Model (Simple Linear Regression)

Independent Variable (X): Day of the year (Time). Dependent Variable (Y): Total Daily Sales. Goal: Determine if sales are significantly increasing or decreasing as the year progresses.

```r
# Model: TotalSales = Intercept + Slope * DayIndex
sales_model <- lm(TotalSales ~ DayIndex, data = daily_sales2)
```

3. Model Evaluation

```r
model_summary <- summary(sales_model)
print(model_summary)
```

```
##
## Call:
## lm(formula = TotalSales ~ DayIndex, data = daily_sales2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -108849  -16947     504   15898   52780
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 70834.64    2341.33   30.25   <2e-16 ***
## DayIndex      127.55      11.06   11.54   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22350 on 364 degrees of freedom
## Multiple R-squared:  0.2677, Adjusted R-squared:  0.2657
## F-statistic: 133.1 on 1 and 364 DF,  p-value: < 2.2e-16
```

Model Evaluation Intercept(70834.64): The model estimates that on Day 0 (the very start of the year), the business would have made about $70834.64 in sales. **DayIndex(127.55)**: For every single day that passes in 2019, the total sales increased by an average of $127.55.Since this number is positive, the business is growing daily throughout the year.

P-Value: The P-Value is extremely small, showing statistically significant. Because the value is less than 0.05, we are confident that the increase in sales is real and not just due to random luck.

R-Squared Value

```r
# Extract R-squared to see how well the date explains sales variance
r_squared <- model_summary$r.squared
cat("R-squared:", r_squared, "\n")
```

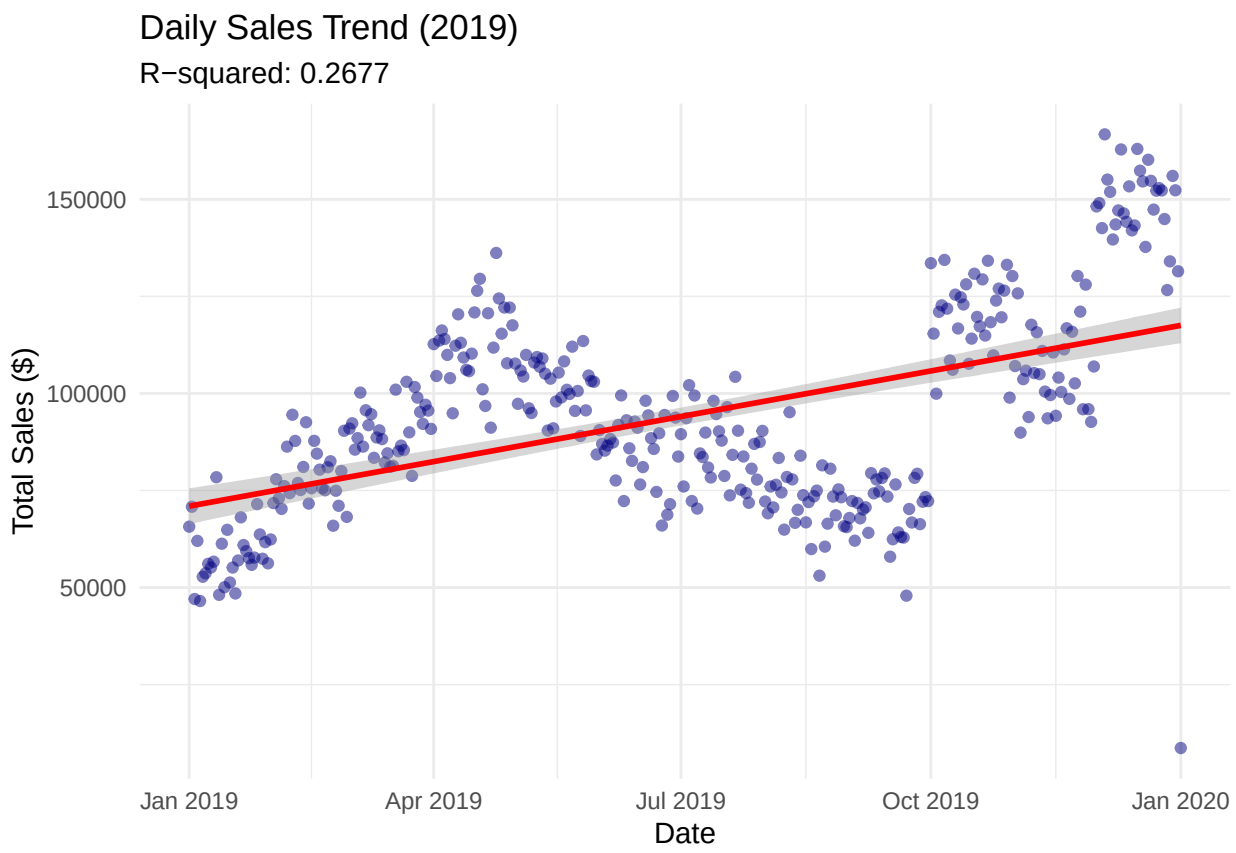
```
## R-squared: 0.2677058
```

While there is a clear upward trend (sales increasing by ~$127/day), the date alone only explains about 27% of the sales volume. The remaining 73% of variation is *likely due to weekly cycles (weekends vs weekdays) and holiday spikes.*

4. Visualization with Regression Line

```r
ggplot(daily_sales2, aes(x = Date, y = TotalSales)) +
  geom_point(alpha = 0.5, color = "navyblue") +      # Scatter plot of actual data
  geom_smooth(method = "lm", color = "red") +    # The regression line
  labs(title = "Daily Sales Trend (2019)",
       subtitle = paste("R-squared:", round(r_squared, 4)),
       x = "Date",
       y = "Total Sales ($)") +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# ggsave("outputs/figures/sales_trend_regression.png")
```